SkillSetu

Bridging Skills to Careers with AI

AI-Based Career Guidance, Skill-Gap Identification and Learning-to-Career Pathways

MVP Implementation Blueprint

Architecture | User Flow | Frontend | Backend | AI Design | UI/UX | Build Plan

Smart India Hackathon 2026

Document version 1.1 | October 2026

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1. Overview

1.1 Problem

Students in India often choose careers with little personalised guidance. They do not know which skills a role needs, how their own skills compare, or what to learn next and in what order. Colleges and training bodies see the same gap at scale but have no simple way to measure it. The result is a mismatch between what students learn and what employers expect.

1.2 Solution

SkillSetu is a web platform (mobile-first) that takes a student from "I don't know what to do" to "here is my target role, my exact skill gaps, and a week-by-week plan to close them". It does this in five steps:

1. **Assess** the student's current skills through a short quiz, self-rating and optional resume upload.
2. **Match** the student to careers using a transparent scoring model, with a plain-language reason for every match.
3. **Identify gaps** between the student's skills and the target role, ranked by importance.
4. **Build a pathway**: an ordered, time-boxed learning plan using free and low-cost courses plus small projects.
5. **Connect** the student to internships, jobs and mentors that fit the skills they have or are building.

> **What makes this different**
>
> Explainable: every recommendation shows why it was made, so students and institutions can trust it.
>
> Practical: the output is a dated roadmap, not just a list of careers.
>
> Inclusive: works on low-end phones and weak networks, and supports English, Hindi and Marathi.
>
> Useful to institutions: cohort dashboards show where students are weak and which skills are in demand.

1.3 India-specific design

SkillSetu is built for learners who get the least career guidance today: ITI, polytechnic and diploma students, and students in tier-2 and tier-3 degree colleges, many of them first-generation learners.

Design choice	What it means in the product
Framework alignment	Every role is tagged with its NSQF level and NCO occupation code, and with the relevant Qualification Pack from the Sector Skill Council where one exists. Mappings are reviewed by a domain expert before use.
Vocational and degree routes	Pathways can include short-term skill programmes, apprenticeships and diploma-to-degree steps, not only online courses.
Stepping-stone careers	If the target role is far away, the plan shows a nearer first role (for example Electrician, then Maintenance Technician, then Electrical Supervisor).

Design choice	What it means in the product
Language and access	English, Hindi and Marathi; OTP login by phone; small app size; works on a 360 px screen over weak networks.
Local demand	Start with demand data for one state (for example Maharashtra) and add more states later, so recommendations reflect jobs students can actually reach.

1.4 How SkillSetu differs from common tools

Common tool type	Typical limitation	SkillSetu approach
Online career tests	Give a generic interest or personality result, not tied to skills or next steps	Combines interests with measured skills and shows concrete gaps
Course platforms	Large catalogues but little help choosing what to learn first	Selects and orders courses from the student's own gaps and weekly time
Job portals	Show openings and assume the student is job-ready	Shows readiness and stepping-stone roles before applying
Placement cells	Mostly manual, little data on skills or outcomes	Cohort analytics plus outcome tracking after training

On top of this, SkillSetu has three signature features that are easy to demonstrate and hard to find together in typical tools: a vocational **Career Ladder**, an **Explain-and-Challenge** mode, and **voice-first guidance** in Hindi and Marathi. They are described in Section 2.

2. What Makes SkillSetu Unique

Career guidance is a crowded space, so SkillSetu is built around three signature features. Each one answers a real gap, each can be shown live in under a minute, and each reuses the same skill and role data, so the extra build effort is small.

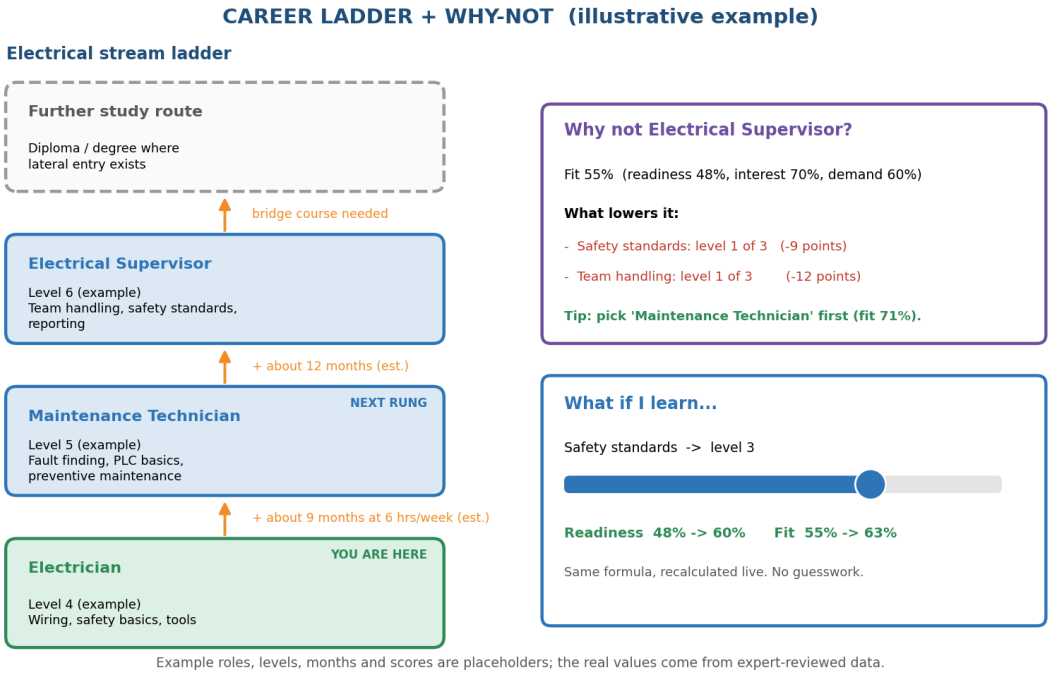


Figure 1: Career Ladder with Why-not and What-if panels (illustrative example)

2.1 Signature feature 1: Vocational Career Ladder

The gap: most career tools stop at one job title and assume a degree. Students from ITI, diploma and skill-training routes rarely see what comes after their first job or how long each step takes.

What it does: for each stream, the app shows a ladder of roles. Every rung shows its NSQF level, the key skills, the bridge skills missing from the rung below, and an estimated time to reach it at the student's weekly study hours. The student's current position and next rung are highlighted, and tapping a rung opens its skill-gap report and pathway.

Item	Design
Data	role_transitions table (from role, to role, typical months, bridge note) plus a stream tag on each role. This extends the stepping-stone idea in Section 6.7.
Bridge skills	Skills required by the next rung that are not required by the current one, computed with the same gap engine.
Time estimate	Sum of the hours of the courses needed for the bridge skills divided by the student's hours per week.
Routes	Upward steps, lateral moves and further-study routes where they exist. Confirm every route and entry rule with experts and current official sources before publishing.

Item	Design
MVP scope	Three streams with four to five rungs each (suggested: electrical, IT / data, manufacturing). Example rungs to validate: Electrician, Maintenance Technician, Electrical Supervisor.
Demo moment	Pick a trade student, show the ladder, tap the next rung and watch the gap report and an 8-week plan appear.

2.2 Signature feature 2: Explain-and-Challenge

The gap: students and institutions are rarely told why a tool suggested something, so they either trust it blindly or ignore it.

What it does: every recommendation can be questioned.

- **Why not this role?** Tap a role the student was not matched to and see its fit score, the skills that lower it, and how many points each costs.
- **What if I learn this?** Move a skill slider and see readiness and fit recalculate live using the same formula.
- **Challenge a result:** a student can flag a recommendation as wrong and say why; flags go to the admin review list and feed the weight calibration in Section 6.8.

Item	Design
How it is computed	Deterministic: recompute the fit formula with the changed skill level and report the difference. No LLM is needed for the numbers; the LLM only phrases the explanation in the student's language.
API	GET /careers/{role_id}/why-not returns the fit breakdown and top lowering skills. POST /simulate takes a list of skill changes and returns new readiness and fit.
UI	"Why not?" button on every role card; slider panel on the gap report; "Report a wrong match" link.
Why it matters	Builds trust, motivates action by showing the payoff of each skill, and supports the fairness story because the logic is visible and testable.
Effort	Low, because the formulas already exist.
Demo moment	Ask "why not Supervisor?", then drag the safety-standards slider and watch the fit rise from 55% to 63%.

2.3 Signature feature 3: Voice-first guidance in Hindi and Marathi (stretch)

The gap: text-heavy apps are hard for many first-generation learners, and typing in regional languages on a phone is slow.

What it does: a microphone button on the home screen and chat lets the student ask in Hindi or Marathi, and the app can read key screens aloud.

- Three voice intents for the MVP: "What is my next step?", "Explain my skill gaps", and "Show careers for me".

- Speech to text uses the browser's speech recognition where the phone supports it, with a cloud speech service as a fallback; text-to-speech reads the next step and the gap summary aloud.
- Text always remains available, and every spoken answer is also shown on screen.

Item	Design
Components	VoiceButton, SpeakButton, an intent router on the backend that maps the transcript to existing APIs.
Risk	Recognition quality for regional accents and browser support for Marathi vary. Test on the phones your students use before promising this in the demo.
Priority	P1 stretch. Cut it first if time runs short; the rest of the product does not depend on it.
Demo moment	Ask in Hindi, "Mera agla kadam kya hai?" (What is my next step?) and hear the answer.

2.4 How the three features fit together

Feature	Problem solved	Effort	Priority	Demo time
Vocational Career Ladder	No visible next steps for ITI and diploma students	Medium (mostly data)	P0 for one stream, P1 for three	50 s
Explain-and-Challenge	Black-box recommendations and low trust	Low	P0	(part of the 50 s)
Voice-first guidance	Text and typing barriers	Medium	P1 stretch	20 s

Parked ideas for later

Proof-based skill verification with auto-checked hands-on tasks.

A portable skill passport that students own and share.

Employer-validated local skill demand by district.

These are listed under Future Scope. One polished signature feature beats three unfinished ones, so build the first two before touching the third.

3. Users and Roles

Role	Who they are	What they need	Main screens
Student	College / ITI / polytechnic student or fresh graduate	Know suitable careers, see skill gaps, follow a learning plan, find opportunities	Onboarding, Assessment, Skill Profile, Careers, Gap Report, Pathway, Opportunities, Chat
Institution / TPO	Training and placement officer, faculty, department head	See cohort readiness, common gaps, and demand trends; export reports	Cohort dashboard, Skill-gap heatmap, Reports
Mentor / Industry	Industry professional or recruiter	Post roles with required skills; see skill-matched students; offer mentoring	Post role, Matched students, Requests
Admin	Platform team	Maintain skills, roles and course data; tune scoring weights	Data management, Weights, Audit log

MVP focus: the Student role is built fully. Institution and Mentor roles get a lean version (one dashboard and one form each). Admin uses seed files and a basic CRUD screen.

4. MVP Scope

The MVP must prove one complete loop: assessment, career match, skill gap, learning pathway. Everything else supports that loop. Priority P0 means "must work in the demo", P1 means "build if time allows".

#	Feature	Priority	MVP approach
1	Sign up / login (email + OTP or password)	P0	JWT auth, role-based access
2	Onboarding (education, interests, language)	P0	3-step form, saved to profile
3	Skill assessment	P0	Quiz of 5 questions per skill cluster plus self-rating; question bank in JSON
4	Skill profile with radar chart	P0	Levels 0-5 per skill; verified vs self-declared flag
5	Career recommendations (top 5) with reasons	P0	Rule-based scoring plus embedding similarity; reasons from template or LLM
6	Skill-gap report for a chosen role	P0	Weighted gap formula; green / orange / red view
7	Learning pathway (weekly roadmap)	P0	Course selection per gap, prerequisite ordering, hours-per-week fit
8	Progress tracking and re-assessment	P0	Mark items complete, mini-quiz updates skill level
9	Opportunities list (internships / jobs)	P1	Seeded listings matched by skills; apply button logs interest
10	AI mentor chat	P1	LLM with the student's profile as context; guardrails
11	Institution dashboard	P1	Cohort readiness, top 10 gaps, export CSV

#	Feature	Priority	MVP approach
12	Mentor / industry portal	P1	Post role with skills; view matched students
13	Resume upload and skill extraction	P1	PDF to text, then skill matching against taxonomy
14	Multilingual UI (EN / HI / MR)	P1	i18n JSON files; LLM answers in chosen language
15	Role tags: NSQF level and NCO code	P0 data	Stored in role data; shown on role detail page
16	Stepping-stone roles	P1	Role transition graph; suggested when readiness is low
17	Outcome follow-up form (opt-in)	P1	Form plus dashboard tile with sample data; scheduler later
18	Weight calibration with expert panel	P0 process	Offline exercise before the pilot (Section 6.8)
19	Vocational Career Ladder (Section 2.1)	P0 / P1	One stream for P0; three streams for P1
20	Why-not and What-if simulator (Section 2.2)	P0	Deterministic recompute of the fit formula
21	Voice-first guidance (Section 2.3)	P1 stretch	Three voice intents in Hindi and Marathi

Out of scope for MVP (listed under Future Scope)

Live integrations with job portals, DigiLocker and course provider APIs (MVP uses seeded files).

Native mobile apps (the PWA covers mobile).

Payments, certificates issued by the platform, video interviews, and advanced proctoring.

5. System Architecture and Tech Stack

The system has four layers. A **modular monolith** is used for the backend: one deployable application with clearly separated modules. This is faster to build and debug than microservices, and each module can be split out later.

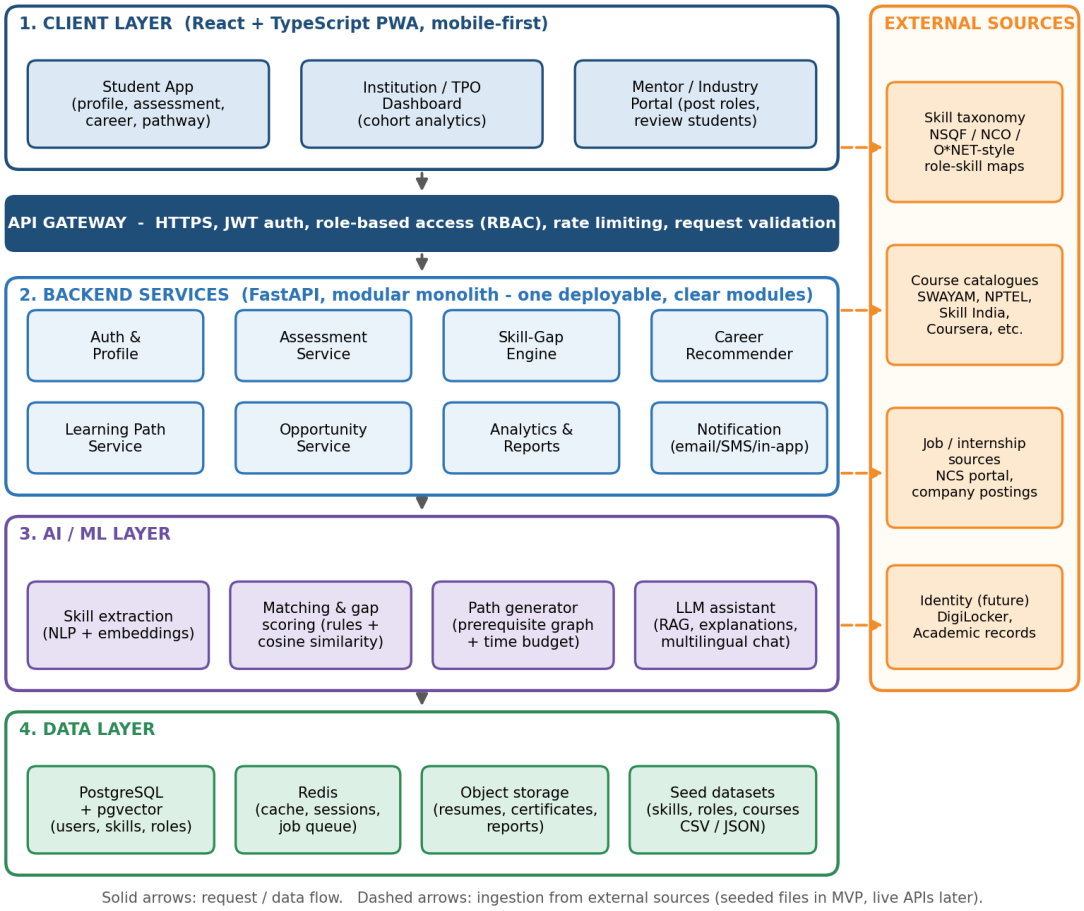


Figure 2: SkillSetu system architecture

5.1 Layer responsibilities

Layer	Responsibility
Client	React PWA for students, institutions and mentors. Calls the backend through REST APIs. Works offline for already-loaded pages.
API gateway	TLS, JWT verification, role checks, rate limiting and input validation before any request reaches a service.
Backend services	Business logic split into modules: auth/profile, assessment, skill-gap, career recommender, learning path, opportunities, analytics, notifications.
AI / ML	Skill extraction and matching (embeddings), scoring engines (rules), pathway generation (graph ordering) and an LLM assistant with retrieval for chat and explanations.
Data	PostgreSQL with pgvector for structured data and embeddings; Redis for cache, sessions and background jobs; object storage for uploads.

Layer	Responsibility
External sources	Skill taxonomies, course catalogues and job sources. In the MVP these are loaded from seed files; live connectors come later.

5.2 Recommended tech stack

Area	Choice	Why
Frontend	React 18, TypeScript, Vite, Tailwind CSS	Fast build, typed code, quick styling, large ecosystem
UI components	shadcn/ui or Material UI, Recharts for charts	Accessible components; radar and bar charts out of the box
State / data	TanStack Query (server data), Zustand (small UI state)	Less boilerplate than Redux; built-in caching
i18n / PWA	react-i18next, Workbox service worker	Multilingual UI; installable and fast on weak networks
Backend	Python 3.11, FastAPI, Pydantic, SQLAlchemy, Alembic	Best fit with ML code; auto-generated API docs; easy validation
Database	PostgreSQL 15 with pgvector	One database for relational data and similarity search
Cache / jobs	Redis with RQ or Celery	Resume parsing and email run in the background
AI / ML	sentence-transformers (multilingual MiniLM), scikit-learn, NetworkX, an LLM API for chat and explanations	Small models run on CPU; LLM used only where language quality matters
Files	S3-compatible storage (MinIO locally)	Resumes and reports
DevOps	Docker Compose, GitHub Actions, deploy on Render / Railway / AWS	One command local setup; simple CI
Monitoring	Sentry, structured logs, health endpoint	Catch errors during the demo period

6. AI / ML Design

The AI is deliberately **hybrid**: transparent scoring rules for decisions that must be explainable, embeddings for fuzzy text matching, and an LLM only for natural-language explanation and chat. This keeps results consistent, auditable and cheap to run.

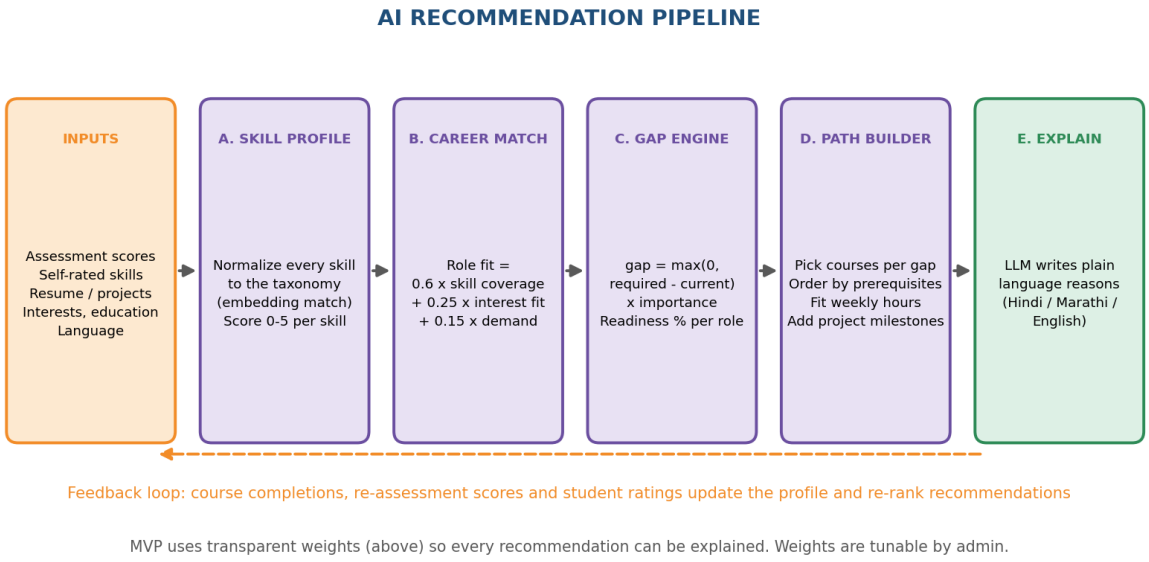


Figure 3: AI recommendation pipeline

6.1 Stage A - Skill profile builder

- **Inputs:** quiz scores, self-ratings (0-5), resume text, projects and certifications.
- **Normalisation:** each free-text skill (for example "MS Excel", "excel formulas") is embedded and matched to the nearest skill in the taxonomy using cosine similarity (accept above 0.75, otherwise ask the student to confirm).
- **Level calculation:** level = 0.7 x quiz score (scaled to 0-5) + 0.3 x self-rating. A self-rated skill without a quiz is capped at level 3 and marked "unverified".
- **Output:** a vector of skill levels plus a verified / unverified flag for each skill.

6.2 Stage B - Career matching

For each role in the catalogue the system computes a **fit score** from 0 to 100:

```
fit = 0.60 x skill_coverage  +  0.25 x interest_fit  +  0.15 x market_demand
```

- **skill_coverage:** the readiness score from Stage C for that role.
- **interest_fit:** cosine similarity between the student's stated interests and the role description embedding.
- **market_demand:** a 0-1 value per role from the demand table (seeded from public job data, updated by admin).
- The top 5 roles are shown. Weights are stored in the database so an admin can tune them without code changes.

6.3 Stage C - Skill-gap engine

Every role has a list of required skills, each with a required level (0-5) and an importance weight (0-1). The engine computes:

```
weighted_gap(skill) = max(0, required - current) x importance
readiness (%)       = sum( min(current, required) x importance )
                      / sum( required x importance )  x 100
```

Worked example: target role Data Analyst

Skill	Required	Current	Importance	Weighted gap	Status
SQL	4	2	0.9	1.80	Partial (priority 1)
Python	3	1	0.8	1.60	Missing (priority 2)
Data visualisation	3	1	0.6	1.20	Missing (priority 3)
Statistics	4	3	0.8	0.80	Partial
Communication	3	2.5	0.5	0.25	Nearly there
Excel	4	4	0.6	0.00	Have

Readiness = 9.25 / 14.9 = **62%**. The gaps are sorted by weighted gap, so the pathway starts with SQL and Python.

6.4 Stage D - Learning pathway builder

- **Candidate courses:** for each gap skill, fetch courses that teach that skill at the needed level range, filtered by the student's language, budget (free-first) and device constraints.
- **Ordering:** build a prerequisite graph (for example SQL basics before advanced SQL; Python before data visualisation with Python) and sort it topologically; within a level, order by weighted gap.
- **Time fit:** the student sets hours per week; the engine packs items into weekly slots based on each course's duration and stops at the target date.
- **Milestones:** after every two weeks a mini-quiz re-assesses the skills; every pathway ends with a small portfolio project.
- **Output:** a pathway with weeks, items (course / project / quiz), estimated hours and the skill each item improves.

6.5 Stage E - Explanations and chat assistant

- The LLM receives only structured facts (scores, gaps, selected courses) and is asked to write a short reason in the student's language. It does not decide rankings.
- The chat assistant answers career questions using retrieval over the platform's own role, skill and course data, plus the student's profile. If the answer is not in the data it says so and suggests asking a mentor.
- Guardrails: no promises about jobs or salaries, no medical or legal advice, a refusal template for off-topic requests, and a daily message cap per user.

6.6 Seed data needed for the MVP

Dataset	Size for MVP	Source ideas
Skills taxonomy	60-80 skills grouped in clusters	Build from O*NET and ESCO (check licences), NSQF qualification packs, NCO occupations
Roles with required skills	15-20 roles across IT, data, core engineering, business, healthcare support	Map from public occupation data; review with a domain expert

Dataset	Size for MVP	Source ideas
Courses	100-150 entries with skill tags, level, hours, language, cost	Public catalogues such as SWAYAM, NPTEL, Skill India; store title, link and metadata only
Question bank	5-8 questions per skill cluster, tagged by difficulty	Write in-house; ask faculty to review
Opportunities	30-50 sample internships and jobs	Sample data for demo; later live connectors

Fairness and quality checks

Do not use gender, caste, religion or location as inputs to scoring.

Test with at least 10 sample student profiles and ask faculty to judge whether the top 5 roles make sense.

Show confidence: if a student has few verified skills, label the matches as "early estimate".

6.7 Stepping-stone roles

A student with 30% readiness for a distant role needs a nearer first step. The platform keeps a small graph of role transitions (from role, to role, typical months). When readiness for the chosen target is below a threshold (default 50%, set by admin), the engine suggests the closest reachable role that has a transition path to the target. The pathway then shows two stages: Step 1 for the nearer role, Step 2 for the target. This turns a single recommendation into a multi-step career pathway.

6.8 Validation and calibration of the scoring weights

The weights in Section 6.2 (0.60, 0.25, 0.15) and the 50% threshold are **starting hypotheses, not proven values**. They are checked and tuned before the pilot:

- **Test profiles:** prepare 30 profiles (a mix of real consented students and realistic synthetic ones) across six streams, such as IT, electrical, mechanical, commerce, healthcare support and arts.
- **Expert labels:** five reviewers (faculty, industry mentors, a career counsellor) independently choose the three most suitable roles for each profile from the role catalogue.
- **Metrics:** top-3 hit rate (system's top 3 contains an expert-chosen role), Spearman rank correlation with the expert consensus, and agreement between the experts themselves as an upper limit.
- **Baselines:** compare against interest-only matching, skill-only matching and random ranking.
- **Tuning:** grid search on 70% of the profiles, test on the remaining 30%, then freeze the weights and record them in the scoring_weights table.

Check	Goal to aim for	How it is reported
Top-3 hit rate	70% or higher on the held-out profiles	Table in the final presentation
Improvement over interest-only baseline	15 percentage points or more	Side-by-side bar chart

Check	Goal to aim for	How it is reported
Expert agreement (ceiling)	Reported as measured, whatever it is	Shows how subjective the task is

These are goals to test, not results already achieved. If the system misses a goal, report the real number and the fix you made.

7. User Flow

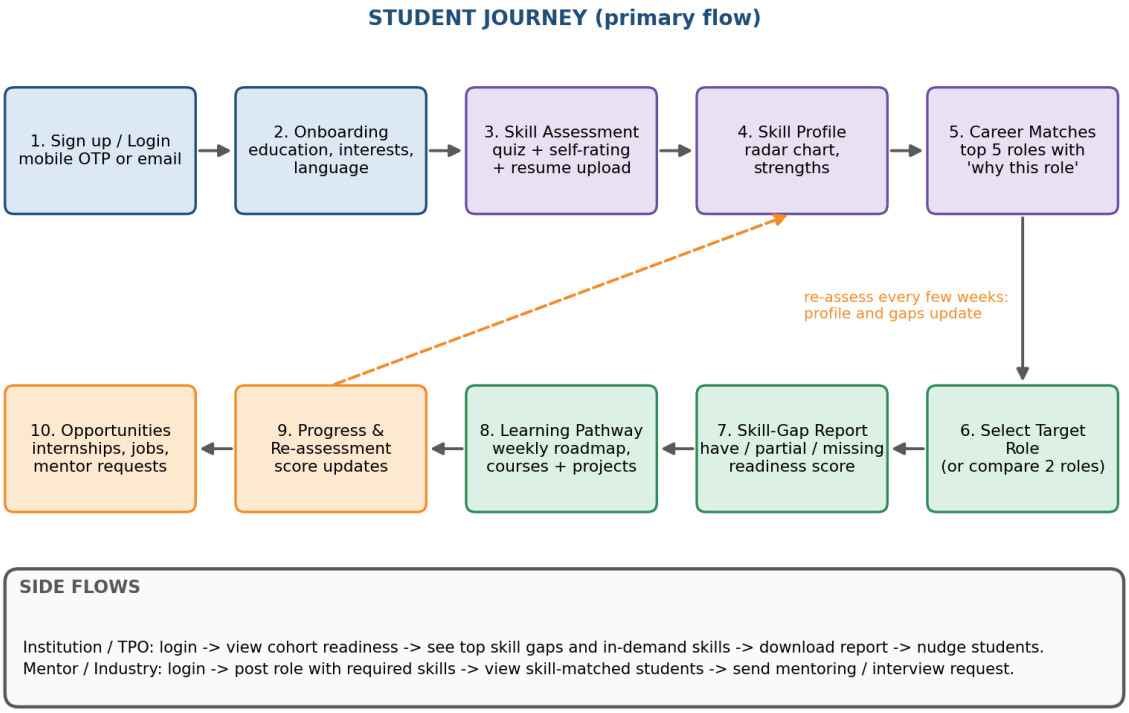


Figure 4: Student journey and side flows

7.1 Student flow in detail

Step	Screen	What the student does	What the system does
1	Login / Sign up	Enters email or phone, verifies OTP	Creates account, issues JWT
2	Onboarding	Picks education level, branch, interests, preferred language	Saves profile; sets UI language
3	Skill assessment	Answers quiz, self-rates skills, optionally uploads resume	Scores answers, extracts skills, stores levels
4	Skill profile	Reviews radar chart and strengths	Shows verified and unverified skills
5	Career matches	Browses top 5 roles, reads reasons, opens a role	Computes fit scores and explanations
6	Choose target role	Selects one role (or compares two)	Saves target role
7	Skill-gap report	Sees readiness %, gaps ranked by priority	Runs gap engine
8	Learning pathway	Sets hours per week and target date, accepts plan	Builds weekly roadmap
9	Progress	Marks items done, takes mini-quiz	Updates skill levels and readiness
10	Opportunities	Views matched internships and jobs, requests mentor	Matches by current skills; logs interest

7.2 Institution flow

1. Login with an institution account (created by admin).
2. Open the **Cohort dashboard**: number of students assessed, average readiness, distribution by target role.
3. Open **Skill-gap heatmap**: top 10 skills missing across the cohort, and which are in demand.
4. Download a CSV or PDF report; send a reminder to students who have not finished the assessment.

7.3 Mentor / industry flow

1. Login and complete a short organisation profile.
2. Post a role: title, required skills with levels, location, type (internship / job).
3. View a ranked list of students whose skills match, with their verified skills only.
4. Send a mentoring or interview request; the student gets an in-app notification.

7.4 Edge cases to handle

- Student skips the assessment: show matches based on interests only and label them "early estimate".
- Student changes target role: keep the old pathway archived and generate a new one.
- No course found for a skill: show a project-based task and flag the gap to admin.
- Weak network: queue progress updates and sync when online.

8. Frontend Design

8.1 Approach

- Single-page app built as a **PWA**: installable, caches static assets, and shows a friendly offline page.
- **Mobile-first** layouts from 360 px wide; desktop layout adds side navigation.
- Role-based routing: after login the user lands on the dashboard for their role.
- Server data through TanStack Query with caching; forms through React Hook Form with Zod validation.

8.2 Pages and routes

Route	Page	Role	Key content
/login, /register	Auth	All	Email / OTP form, language switcher
/onboarding	Onboarding wizard	Student	3 steps: education, interests, language
/assessment	Skill assessment	Student	Cluster picker, quiz runner, self-rating sliders, resume upload
/profile	Skill profile	Student	Radar chart, skill list with verified badges
/careers	Career matches	Student	Role cards with fit %, reason, demand tag, compare
/careers/:id	Role detail	Student	Required skills, typical tasks, salary band range, button: Set as target
/ladder/:stream	Career ladder	Student	Rungs with NSQF level, bridge skills, time estimate, you-are-here marker
/gap/:roleId	Skill-gap report	Student	Readiness ring, coloured bars, ranked gap list, what-if sliders
/pathway	Learning pathway	Student	Timeline by week, item status, mini-quiz, progress bar
/opportunities	Opportunities	Student	Filter by role and type, apply / save
/chat	AI mentor	Student	Chat with suggested prompts
/institution	Cohort dashboard	Institution	KPIs, heatmap, export
/mentor	Mentor portal	Mentor	Post role, matched students, requests
/admin	Admin data	Admin	CRUD for skills, roles, courses; weights

8.3 Reusable components

Component	Purpose
AppShell, TopBar, BottomNav, SideNav	Layout; bottom navigation on mobile, side navigation on desktop
ProtectedRoute	Redirects by login state and role

Component	Purpose
StepWizard	Multi-step forms with progress dots
QuizRunner / QuestionCard	One question per screen, timer optional, saves each answer
SkillSlider	0-5 self-rating with labels (none, basic, working, strong, expert)
RadarChart / SkillBar	Skill visualisation (Recharts)
ReadinessRing	Circular progress for readiness %
RoleCard	Role title, fit %, why-this-role text, demand tag
GapList	Ranked gaps with colour status and "Fix this" button
CareerLadder / LadderRung	Vertical ladder of roles with status, level and time between rungs
WhyNotPanel	Fit breakdown and the skills that lower it
WhatIfSlider	Change a skill level and see readiness and fit update live
VoiceButton / SpeakButton	Speech input and read-aloud with text fallback (stretch)
PathwayTimeline / PathwayItem	Weekly timeline; status: locked, active, done
OpportunityCard	Title, company, skills matched, apply
ChatPanel	Message list, input, suggested questions, language-aware
EmptyState, ErrorState, Skeleton	Consistent loading, empty and error handling
LanguageSwitcher, Toast, Modal	Shared UI utilities

8.4 Suggested folder structure

```
frontend/
  src/
    app/              routes.tsx, providers.tsx, queryClient.ts
    components/       ui/ (buttons, inputs), charts/, layout/
    features/
      auth/           LoginPage, RegisterPage, useAuth
      onboarding/     OnboardingWizard
      assessment/     QuizRunner, SelfRating, ResumeUpload
      profile/        SkillProfilePage
      careers/        CareerMatchesPage, RoleDetailPage
      gap/            GapReportPage
      pathway/        PathwayPage, MiniQuiz
      opportunities/
      chat/
      institution/    CohortDashboard
      admin/
    lib/              api.ts (axios), i18n.ts, validators.ts
    locales/          en.json, hi.json, mr.json
    styles/           tailwind.css, tokens.css
  public/             manifest.webmanifest, icons/
```

8.5 Performance and accessibility targets

- First load under 3 seconds on a mid-range phone over 4G; initial JS bundle under 250 KB (route-level code splitting).
- Lighthouse scores of 90+ for performance and accessibility.

- Keyboard navigable, visible focus states, labels on every input, alt text on charts with a text summary.

9. Backend Design and APIs

9.1 Module structure

```
backend/
  app/
    main.py              FastAPI app, middleware, routers
    core/                config, security (JWT, hashing), deps
    db/                  models.py, session.py, migrations/ (Alembic)
    modules/
      auth/              router, service, schemas
      profile/
      assessment/        question bank, scoring
      skills/            taxonomy, embedding matcher
      careers/           recommender (fit score)
      gap/               gap engine
      pathway/           course selector, graph ordering, scheduler
      opportunities/
      institution/       analytics queries
      chat/              LLM client, retrieval, guardrails
      admin/
    workers/             resume_parser.py, notifications.py
  data/seed/             skills.csv, roles.csv, role_skills.csv, courses.csv,
questions.json
  tests/
```

9.2 REST API

All endpoints are under /api/v1, return JSON, and require a bearer token unless marked public.

Method	Endpoint	Purpose
POST	/auth/register (public)	Create account with role
POST	/auth/login (public)	Password or OTP login; returns access and refresh tokens
GET / PUT	/me, /me/profile	Read and update profile (education, interests, language)
GET	/assessments/questions?cluster=	Get quiz questions for a skill cluster
POST	/assessments/submit	Submit answers and self-ratings; returns updated skill levels
POST	/resume/upload	Upload PDF; queues parsing; returns job id
GET	/skills/profile	Current skill vector with verified flags
GET	/careers/recommendations	Top roles with fit score, readiness, reasons
GET	/careers/{role_id}	Role detail with required skills
GET	/careers/{role_id}/steps	Stepping-stone roles toward the target
GET	/careers/ladder?stream=	Ladder rungs for a stream with NSQF level, bridge skills and time estimates
GET	/careers/{role_id}/why-not	Fit breakdown and the skills that lower it

Method	Endpoint	Purpose
POST	/simulate	Body: skill changes. Returns new readiness and fit (what-if)
POST	/voice/intent	Transcript in, matched intent and answer out (stretch)
POST	/targets	Set target role for the student
POST	/gap/analyze	Body: role_id. Returns readiness and ranked gaps
POST	/pathways	Body: role_id, hours_per_week, target_date. Returns pathway
GET	/pathways/{id}	Pathway with weekly items and status
PATCH	/pathways/{id}/items/{item_id}	Mark item started / done; may trigger mini-quiz
GET	/opportunities?role_id=&type=	Matched opportunities
POST	/opportunities (mentor)	Create a role posting
POST	/opportunities/{id}/apply	Log interest or application
POST	/chat	Send message; returns assistant reply (rate limited)
GET	/institution/summary	Cohort KPIs
GET	/institution/skill-gaps	Top gaps across cohort
GET	/institution/export?format=csv	Download cohort report
GET / POST	/outcomes/due, /outcomes	Pending follow-ups; submit an outcome report (opt-in)
GET	/institution/outcomes	Aggregated outcomes by course and cohort
CRUD	/admin/skills, /roles, /courses, /weights	Manage data and scoring weights

9.3 Sample request and response

POST /api/v1/gap/analyze

```
Request:
{ "role_id": "data_analyst" }

Response:
{
  "role": "Data Analyst",
  "readiness": 62,
  "gaps": [
    { "skill": "SQL", "required": 4, "current": 2,
      "weighted_gap": 1.8, "status": "partial" },
    { "skill": "Python", "required": 3, "current": 1,
      "weighted_gap": 1.6, "status": "missing" }
  ],
  "strengths": ["Excel", "Communication"],
  "explanation": "You work well with spreadsheets. SQL and
                  Python are the fastest way to this role."
}
```

9.4 Background jobs

- **Resume parsing:** extract text, find skills, propose matches for the student to confirm.
- **Notifications:** reminders for pending assessments and weekly pathway nudges (email, in-app; SMS later).
- **Nightly analytics:** refresh cohort summary tables and demand scores.

9.5 Error handling and conventions

- Standard error shape: { "error": { "code", "message", "details" } } with proper HTTP status codes.
- Request validation with Pydantic; pagination with limit and cursor; idempotent PATCH for progress updates.
- OpenAPI documentation auto-generated at /docs for the frontend team.

10. Database Design

PostgreSQL with the pgvector extension. Main tables and key columns are listed below. All tables have id, created_at and updated_at.

Table	Key columns	Notes
users	email, phone, password_hash, role, language, institution_id	Role: student, institution, mentor, admin
student_profiles	user_id, education_level, branch, year, interests[], target_role_id	One per student
institutions	name, type, city, state	Links students to a college
skills	name, cluster, description, embedding vector(384)	Taxonomy; embedding for matching
roles	title, description, stream, demand_score, nsqf_level, nco_code, qp_code, embedding vector(384)	Career catalogue; NSQF / NCO tags
role_skills	role_id, skill_id, required_level, importance	Drives gap engine
user_skills	user_id, skill_id, level, source, verified, updated_at	source: quiz, self, resume, course
question_bank	skill_id, cluster, question, options, answer, difficulty	For assessments
assessment_attempts	user_id, cluster, score, answers_json, taken_at	History for progress charts
courses	title, provider, url, hours, language, cost, level	Metadata only; link out to provider
course_skills	course_id, skill_id, level_from, level_to	What each course teaches
skill_prerequisites	skill_id, prerequisite_skill_id	Graph edges for ordering
pathways	user_id, role_id, hours_per_week, target_date, status	One active per student
pathway_items	pathway_id, week_no, type, ref_id, hours, status, completed_at	Type: course, project, quiz
opportunities	title, company, type, location, required_skills[], posted_by	Internships and jobs
applications	opportunity_id, user_id, status	Interest log
chat_messages	user_id, role, content, created_at	Retain 30 days; user can delete
role_transitions	from_role_id, to_role_id, typical_months, bridge_note	Career ladder and stepping-stone graph
follow_ups	user_id, due_at, channel, status	Scheduled outcome check-ins (opt-in)
outcomes	user_id, status, role_id, sector, employer_type, wage_band, retained_months, source, reported_at	Self-reported; employer-verified later
scoring_weights	key, value	Tunable weights for fit score

Table	Key columns	Notes
audit_log	actor_id, action, entity, entity_id, at	Admin and data-change tracking

Key relationships: users 1-1 student_profiles; roles many-to-many skills through role_skills; users many-to-many skills through user_skills; courses many-to-many skills through course_skills; pathways 1-many pathway_items.

Indexes: user_skills(user_id), role_skills(role_id), pathway_items(pathway_id, week_no), and an ivfflat or HNSW index on skills.embedding for similarity search.

11. UI / UX Design

11.1 Design principles

- **One clear next step.** Every screen has one primary button that moves the student forward.
- **Show, don't tell.** Use charts, colour status and progress rings before paragraphs of text.
- **Explain the why.** Each recommendation has a one-line reason in plain language.
- **Respect low bandwidth.** Skeleton loaders, small images, no heavy animation.
- **Encourage, never judge.** Gaps are shown as "to learn", not "failed"; celebrate small wins.

11.2 Visual style

Element	Specification
Primary colour	Deep blue #1F4E79 (trust, education) for headers and primary buttons
Accent	Saffron #F28C28 for highlights and calls to action; keep use limited
Status colours	Green #2E8B57 = have, orange #F28C28 = partial, red #C0392B = missing. Always paired with icon or text, not colour alone
Neutrals	Text #1A1A1A, secondary #595959, background #FFFFFF and #F7F9FC
Typography	Inter or Noto Sans (supports Devanagari for Hindi and Marathi); body 16 px, headings 20-28 px
Shape and spacing	8 px grid; 12 px rounded cards; minimum touch target 44 x 44 px
Icons	Lucide icon set, consistent stroke width
Dark mode	Optional for P1; define colour tokens from the start so it is easy to add

11.3 Key screen wireframes



Figure 5: Mobile wireframes - student home, skill-gap report and learning pathway

11.4 Screen specifications

Screen	Layout and behaviour
Home	Greeting, readiness ring for the target role, one "Next step" card, top 2-3 career matches, bottom navigation (Home, Skills, Careers, Path, Chat).
Assessment	Cluster picker as cards. Quiz shows one question per screen with progress bar; auto-save each answer; "Skip" allowed. Self-rating uses sliders with text labels.
Skill profile	Radar chart on top (max 8 axes), below it a list grouped by cluster with a verified badge or "Verify with quiz" link.
Career matches	Cards sorted by fit. Each has title, fit %, demand tag, one-line reason, and "View role". A compare button allows two roles side by side.
Gap report	Readiness ring, then ranked list: skill name, bar from current to required level, colour status, and "Add to pathway".
Pathway	Vertical timeline by week. Active item expanded with course link, hours, and a "Mark done" button. Locked items greyed out with the prerequisite shown.
Institution dashboard	Four KPI tiles (students assessed, average readiness, top role, completion rate), a skill-gap heatmap, and a table with filters and export.

11.5 Accessibility and inclusion

- Follow WCAG 2.1 AA: contrast ratio of at least 4.5:1, focus rings, semantic HTML and ARIA labels on charts.

- Language switch available on every screen; all UI strings come from locale files; Devanagari fonts bundled.
- Plain, short sentences at roughly grade 8 reading level; avoid jargon, explain terms on tap.
- Support text-size scaling to 200% without layout breakage.
- Voice input for chat is a future option; keep the chat composer compatible with the keyboard dictation of the phone.

11.6 Microcopy examples

Situation	Text
Empty pathway	You haven't picked a target role yet. Choose a role to see your learning plan.
Low confidence match	Early estimate. Take the 10-minute skill quiz to make this more accurate.
Gap item	SQL: you're at level 2 of 4. About 12 hours of practice will get you there.
Completed week	Week 2 done. Your readiness went up from 62% to 68%.

12. Outcome Tracking and Feedback Loop

Guidance is only useful if it leads to real outcomes. This module follows students after they finish a pathway, with their consent, and feeds the results back into the platform. It also links SkillSetu to the wider goal of tracking employment, apprenticeship and self-employment outcomes after skilling.

12.1 Flow

1. Student completes (or pauses) a pathway; the app asks for **separate, optional consent** to follow up.
2. Follow-up reminders at 3, 6 and 12 months by in-app message, email or SMS.
3. A short outcome form: status (placed, internship, apprenticeship, higher studies, self-employed, still searching), role and sector, employer type, optional wage band, months retained, and which gaps mattered most.
4. Responses are stored in the outcomes table and linked to the student's skill history.
5. Institution and admin dashboards show aggregated results; individual answers are never shown to institutions without the student's permission.

12.2 How outcomes improve the product

Use	Effect
Better recommendations	Observed role transitions update the role graph; roles that real students from similar profiles reach get higher confidence
Demand signals	Actual hiring sectors and roles refine the demand score used in the fit formula
Institution insight	Placement, internship and apprenticeship rates by course and cohort; self-reported reasons for non-placement
Impact evidence	Before-and-after readiness plus outcome rates give evidence for funders and policymakers

> **Limits to state honestly**
>
> Outcomes are self-reported in the MVP; employer verification is a future step.
>
> Response rates will be below 100%, so dashboards must show the number of responses next to every percentage.
>
> Wage data is optional and shown only as bands, never as individual values.

13. Security and Privacy

Area	Measures
Authentication	Password hashing with bcrypt or argon2, short-lived access tokens, refresh token rotation, OTP rate limits
Authorisation	Role-based access control on every endpoint; students can only read their own data; institutions see only their own cohort
Data protection	TLS everywhere, encryption at rest for database and storage, secrets in environment variables or a secrets manager
Privacy and consent	Clear consent screen at sign-up; institutions see aggregated data unless the student opts in to share; mentors see verified skills only
Data minimisation	Collect only what scoring needs; do not collect caste, religion or exact address; allow account and chat deletion
Compliance	Design in line with India's Digital Personal Data Protection Act, 2023 (confirm obligations with a legal advisor before launch)
Application security	Input validation, parameterised queries, file type and size checks for uploads, malware scan hook, CORS allow-list
LLM safety	No personal identifiers sent to the LLM beyond what is needed; prompt-injection filtering on resume text; output length and topic limits
Operations	Audit log for admin changes, daily backups, dependency scanning in CI

14. Implementation Plan

14.1 Suggested team (4 members)

Member	Owns
Frontend developer	React app, components, charts, i18n, PWA
Backend developer	FastAPI modules, database, auth, APIs, deployment
AI / data developer	Skill taxonomy, embeddings, gap engine, pathway builder, LLM chat
UI/UX and content lead	Wireframes, visual design, question bank, role and course data, testing, demo script

14.2 Six-week MVP timeline

Week	Goals	Deliverable
1	Finalise scope; wireframes; set up repo, Docker, CI; database schema; seed data collection starts	Clickable wireframes; running skeleton app; schema migrated
2	Auth and onboarding; skill taxonomy and role data loaded; assessment engine and question bank	Student can sign up and complete a quiz
3	Skill profile and radar chart; career recommender with fit score and reasons; Why-not explanations	Top 5 careers with explanations

Week	Goals	Deliverable
4	Gap engine and report; learning pathway builder; progress tracking; career ladder for one stream; what-if simulator	Full core loop working end to end
5	Opportunities, institution dashboard, chat assistant, i18n (Hindi / Marathi), stepping-stone roles, outcome form; expert-panel weight calibration	P1 features integrated; calibrated weights
6	Mini-pilot with 15-20 students (Section 16), bug fixes, performance, security review, demo prep and deployment	Deployed MVP, calibration and pilot numbers, demo video, documentation

14.3 Hackathon cut (36-48 hours)

If time is limited to a hackathon finals window, build only the core loop and fake the rest with seed data:

- Hour 0-4: repo setup, seed CSVs for 10 roles, 40 skills, 60 courses; login can be a simple demo user.
- Hour 4-14: assessment quiz, skill profile, fit score and top 5 careers.
- Hour 14-24: gap report, pathway builder, progress screen, and the ladder view for one stream with the Why-not panel.
- Hour 24-32: UI polish, LLM explanations, one institution dashboard screen with sample data.
- Hour 32-36: deploy, rehearse the demo, record a backup video, prepare answers on privacy and bias (Section 19), and put any calibration or pilot numbers on one slide, clearly labelled.

14.4 Definition of done for the MVP

- A new student can complete onboarding, assessment, pick a role, see gaps and get a pathway in under 15 minutes.
- Every recommendation shows a reason, and the numbers can be reproduced from the formulas in Section 6.
- The app works on a 360 px screen over a throttled 3G test connection.
- No critical security issues; role-based access verified by tests.

15. Testing and Deployment

15.1 Testing

Type	What	Tool
Unit tests	Gap formula, fit score, level calculation, prerequisite ordering, scheduler	pytest
API tests	Auth, role access, every endpoint happy and failure paths	pytest + httpx
Frontend tests	Components and key flows	Vitest + React Testing Library
End-to-end	Signup to pathway for a sample student	Playwright
Recommendation quality	10-20 sample profiles judged by faculty; track precision of top 5	Manual review sheet
Usability	5-8 students try the flow; measure time to complete and note confusion points	Moderated sessions
Performance and accessibility	Lighthouse, axe checks, throttled-network test	Lighthouse CI

15.2 Deployment

- **Environments:** local (Docker Compose), staging, production.
- **CI/CD:** GitHub Actions runs tests and linting; merge to main deploys to staging; manual approval for production.
- **Hosting for MVP:** frontend on Vercel or Netlify, backend and database on Render, Railway or a small AWS setup (one app container, one PostgreSQL, one Redis).
- **Config:** environment variables for database URL, JWT secret, LLM key, storage credentials.
- **Observability:** Sentry for errors, uptime check on /health, basic request logs and a usage dashboard.

16. Pilot and Evaluation Plan

A small, honest pilot gives the MVP real evidence. The plan below is the full version; for SIH finals run a shorter mini-pilot (15-20 students, 1-2 weeks) and present the numbers as early pilot results.

Item	Plan
Goal	Test whether students understand the guidance, find it useful, and improve measured skills while following a pathway
Participants	30-50 students aged 18 or above from two institutions (for example one polytechnic or ITI and one degree college), with written consent and institution permission
Duration	4 weeks: onboarding and baseline in week 0, pathway use in weeks 1-4, re-assessment and survey at the end
Baseline and re-test	Two parallel quiz forms (A and B) built from the same blueprint with different questions, so the second test does not simply repeat the first
Comparison	If possible, a small comparison group that receives only a standard course list; otherwise report as a single-group pilot
Qualitative feedback	3-5 faculty or counsellor interviews and open comments from students

16.1 What to measure

Measure	Method	Goal to aim for
Skill improvement	Paired comparison of form A and form B scores on the targeted skills	Positive average change; report effect size and sample size
Usability	System Usability Scale (10 questions)	Score of 68 or higher (the commonly cited average)
Relevance of career matches	Student rating of top 5 roles (1-5)	Average of 3.8 or higher
Engagement	Share finishing assessment, generating a pathway, completing 2+ items	As in Section 17.2
Faculty usefulness	Short interview and 5-point rating	Qualitative themes plus average rating

16.2 Ethics and data handling

- Only adults take part; no academic marks or penalties depend on participation.
- Collect the minimum data, store it securely, report results in aggregate, and delete a participant's data on request.
- Tell participants the recommendations are guidance, not a guarantee of jobs.

16.3 Known limits

- Small sample, short duration and self-selection mean results are indicative, not conclusive.
- Without a control group, gains may partly reflect practice or motivation; say so in the report.

- Placement outcomes cannot be measured in 4 weeks; that is what the follow-up module in Section 12 is for.

17. Demo Script and Success Metrics

17.1 Four-minute demo script

1. **Problem (25 s):** show the mismatch between what students learn and what roles need.
2. **Student Riya signs up (25 s):** quick onboarding, language switched to Hindi to show multilingual support.
3. **Assessment (30 s):** short quiz; skill profile radar appears.
4. **Career matches (30 s):** Data Analyst 78%, with a plain-language reason.
5. **Ladder and Why-not (50 s):** open the vocational ladder, ask "why not Supervisor?", drag a skill slider and watch the fit rise.
6. **Gap report (30 s):** readiness 62%, SQL and Python are top gaps.
7. **Pathway (30 s):** 8-week plan at 6 hours a week; mark an item done; readiness rises.
8. **Institution view (20 s):** cohort dashboard with top missing skills and demand.
9. **Close (10 s):** impact and next steps; the voice demo can replace one step if time allows.

17.2 Success metrics

Metric	Target for pilot
Assessment completion rate	70% or more of students who start
Students who pick a target role	60% or more
Pathway acceptance	50% or more generate a pathway
Pathway engagement	40% complete at least 2 items in 4 weeks
Readiness improvement	Average of +8 points after 4 weeks
Recommendation usefulness	75% or more thumbs-up on career matches
Institution adoption	At least 2 institutions viewing the dashboard weekly

All targets are goals to test in the pilot, not results already achieved.

18. Risks and Mitigations

Risk	Impact	Mitigation
Role and skill data quality is weak	Poor recommendations	Start with a small, well-reviewed set; faculty and industry review; admin tools to correct data
Cold start: new students have no skill data	Low-confidence matches	Short adaptive quiz; interest-based matches labelled as early estimates
LLM gives wrong or risky advice	Loss of trust	LLM only explains structured results; retrieval from own data; refusal rules; show a disclaimer
Bias in recommendations	Unfair guidance	No sensitive attributes as inputs; review sample outputs for patterns; transparent weights

Risk	Impact	Mitigation
Low connectivity or low-end devices	Students drop off	PWA, small bundles, save-as-you-go, offline-friendly pages
Privacy concerns from institutions or parents	Low adoption	Explicit consent, aggregated institution view by default, clear data-deletion option
Course links change or disappear	Broken pathways	Weekly link checker; admin flag for broken items; fallbacks
Scope creep during development	Missed demo	Freeze P0 list in week 1; add P1 only after the core loop works

19. Questions Judges May Ask

Question	Suggested answer
Why not just use a general AI chatbot?	A chatbot gives unstructured, hard-to-repeat advice. SkillSetu uses measured skills, fixed and explainable formulas, curated courses, progress tracking and institutional analytics. The LLM only explains results and answers questions from the platform's own data.
How do you avoid biased or wrong guidance?	No sensitive attributes are used in scoring; weights are visible and validated against an expert panel (Section 6.8); low-confidence results are labelled as early estimates; students can ask a human mentor.
Where does the role and skill data come from?	Open taxonomies and public occupation data, aligned to NSQF and NCO, reviewed by faculty and industry experts. Admin tools and Sector Skill Council partnerships keep it current.
How do you trust the quiz results?	Faculty-reviewed question bank with difficulty tags, randomised items, parallel forms for re-testing, and a clear split between verified and self-declared skills.
Will it work on poor networks and cheap phones?	Yes by design: PWA, small bundles, offline-friendly pages, queued progress updates and OTP login.
How do you measure real impact?	Pilot measures (Section 16) for skill change and usability, and the outcome tracking module (Section 12) for placement, apprenticeship and self-employment over 3, 6 and 12 months.
What about privacy?	Consent-based, minimal data, aggregated institution views by default, deletion on request, and design aligned with India's data protection law (to be confirmed with a legal advisor).
How can it be sustained?	Options include institutional licences, state skill-mission partnerships and industry-sponsored learning programmes, while keeping the student experience free.
What happens when the demo data runs out?	The MVP uses seeded data on purpose. The architecture separates data from logic, so live course and job feeds can be added without redesign.

20. Future Scope

- Live integrations with course platforms, job portals (for example the National Career Service portal) and DigiLocker for verified academic records.

- Proof-based skill verification with auto-checked hands-on tasks, and a portable skill passport that students own and share.
- Employer-validated local skill demand by district, feeding the role requirements and the ladder.
- Adaptive assessments that choose the next question based on previous answers.
- Labour-market intelligence: automatic demand updates from job postings by district.
- Native Android app, voice-first guidance and more Indian languages.
- Mentor marketplace with scheduling and feedback.
- Employer-verified outcome data and automated follow-ups at scale, building on the opt-in module in Section 12.

Summary

SkillSetu delivers one complete, explainable loop: assess skills, match careers, find gaps, and build a learning pathway.

Its signature features are the vocational Career Ladder, Explain-and-Challenge, and voice-first guidance in Hindi and Marathi.

The MVP is realistic for a small team: a React PWA, a FastAPI modular monolith, PostgreSQL with pgvector, and a hybrid rules + embeddings + LLM approach.

Institutions gain a live view of cohort readiness, and students get a clear next step every time they open the app.